

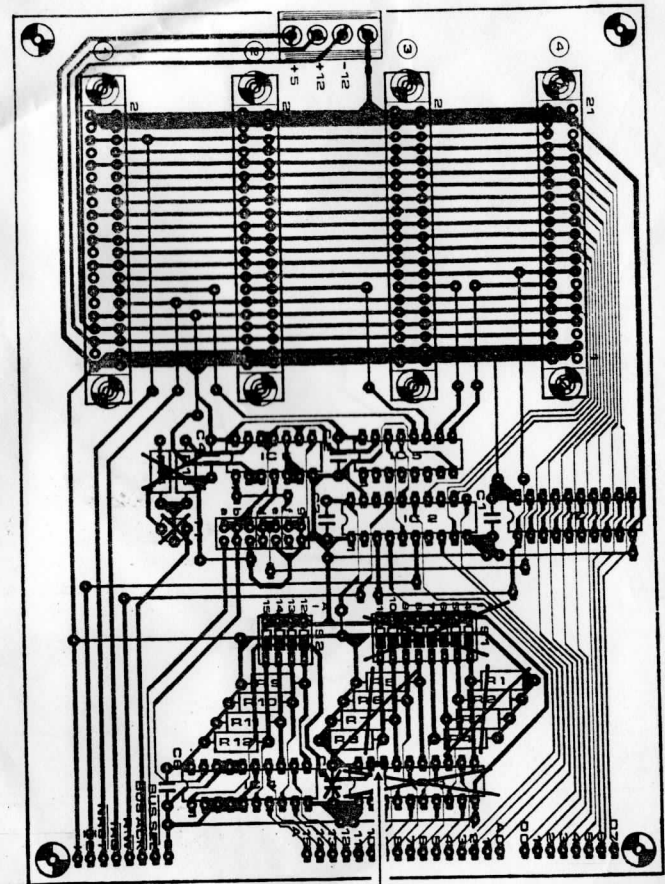


Fig. 2. Modified PCB layout for the I/O bus. Remember to connect pin 18 to pin 19 of the empty IC₃ socket.

modified for use with the Z80 $\overline{WR}$ signal. As for the former, Fig. 3 shows these modifications in heavy lines. IC₂ is now clocked direct by $\overline{WR}$. To make this possible, pin 6 of IC₂ is cut off from the IC body. A wire is installed from IC₂ pin 7 to pin 6 of the bus connector — see Fig. 4. The 8-bit I/O port is modified as follows: disconnect pin 8 of IC₁ by cutting it off. Fig. 6 shows how socket pin 8 of IC₁ is connected to bus pin 6; do not forget to fit the marked wire link on the component

side. Fig. 5 shows the relevant part of the circuit diagram for reference purposes. This completes the adaptations of the digitizer and the 8-bit I/O port for MSX applications.

Ribbon cable connection

Depending on the make of MSX computer, it either features a so-called cartridge slot, a 50-way I/O connector, or both. Connectors for

the cartridge slot are hard to obtain, however, because they are in fact no more than a projecting piece of the PCB on which the expansion circuit (EPROM, interface) is mounted. Cartridges usually come in a small plastic housing, and a cheap one may be salvaged and carefully taken apart, so that it will still function after the flat ribbon cable is connected to it for signal extension towards the I/O bus. A means for switching off the cartridge (i.e. the contained PROM) will be needed when the wires to the I/O bus are connected

SLOT		—	SIGNAL DESCRIPTION
PIN NO.	NAME	I/O	DESCRIPTION
1	CS1	0	ROM 4000 ~ 7FFF select signal (128K)
2	CS2	0	ROM 8000 ~ BFFF select signal (128K)
3	CS12	0	ROM 4000 ~ BFFF select signal (256K)
4	SLTSL	0	Slot selected signal. — Fixed select signal for each slot.
5		—	Reserved for future use only
6	RFSH	0	Refresh signal
7	WAIT	1	Wait signal to CPU (wired-OR)
8	INT	1	Interrupt request signal
9	M1	0	Fetch cycle signal of CPU
10	BUSDIR	1	This signal controls the direction of external data bus buffer when the cartridge is selected. It is low level when the data is sent by the cartridge.
11	IORQ	0	I/O request signal
12	MREQ	0	Memory request signal
13	WR	0	Write signal
14	RD	0	Read signal
15	RESET	0	System reset signal
16		—	Reserved for future use only
17	A9	0	Address bus
18	A15	0	
19	A11	0	
20	A10	0	
21	A7	0	
22	A6	0	
23	A12	0	
24	A8	0	
25	A14	0	Data bus
26	A13	0	
27	A1	0	
28	A0	0	
29	A3	0	
30	A2	0	
31	A5	0	
32	A4	0	
33	D1	I/O	Data bus
34	D0	I/O	
35	D3	I/O	
36	D2	I/O	
37	D5	I/O	
38	D4	I/O	
39	D7	I/O	
40	D6	I/O	
41	GND	—	Ground
42	CLOCK	0	CPU clock 3.58 MHz
43	GND	—	Ground
44,46	SW1,SW2	—	Insert/remove protection, if fitted
45,47	+5 V	—	+5 V power supply
48	+12 V	—	+12 V power supply
49	SOUND IN	1	Sound input (—5 dbm)
50	—12 V	—	—12 V power supply

Input and output is measured with respect to MSX computer

I/O-TEST - 1

```
10 FOR A=0 TO 255
20 IF INP(A)<>255 GOTO 30 ELSE 40
30 PRINTA,INP(A)
40 NEXT
```

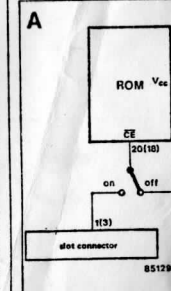
I/O-TEST - 2

```
10 FOR A=0 TO 7
20 OUT112,A
30 PRINTA,INP(112)
40 NEXT
```

in accordance with Table 1. Once the cartridge PCB has been removed (which may require quite an effort depending on the type used), it is necessary to establish the order of the slot connections. The even numbered connections are at the front (label) side of the PCB, the odd ones at the reverse side. With the PCB facing downwards (label side visible), connection 2 is at the utmost right of the front row. The row of odd numbers starts with 1, opposite to 2. To make things absolutely certain: the notch at the rear of the cartridge must now be at the lower left side. The through-plated holes in the cartridge PCB are used to connect the flat ribbon cable to. It may be necessary to first open these holes using desoldering braid. In case there is no through-plated hole available for a connection, the wire end will have to go direct to the appropriate PROM pin. It is useful to use some wires in parallel for the power supply connections, so that 29 of them are required in total. Finally, a feed-through slot is sawed into the plastic box to lead the ribbon cable through.

The above mentioned 50-way I/O connector is definitely much neater for the ribbon cable connection, but special tools are required to successfully fix the (expensive) connector to the cable. Effective disabling of the (EPROM) in the cartridge is not difficult, as it is very likely compatible with the well-known 27XX types. In the case of a 28-pin type, cut the track to pin 20 (CE) and solder a wire from this pin to the pole of a small change-over switch. One of the switch contacts goes to slot connector pin 1 or 3 (+5 V), the other to the track which used to lead to (EPROM) pin 20. In the case of a 24-pin (EPROM), do the same with pin 18 (CE). To disable the cartridge you may now switch its (EPROM) CE pin to +5 V (Fig. A). Table 1 shows the pin designations and signal definitions of the 50-way slot connector. Note that pins 44 and 46 are used for optional computer-

Table 1. Pin designations of the available signals on the MSX 50-way slot connector.



Listing 1. This tiny program checks the connection between MSX computer and the I/O bus, plus the correct function of the modified digitizer board.

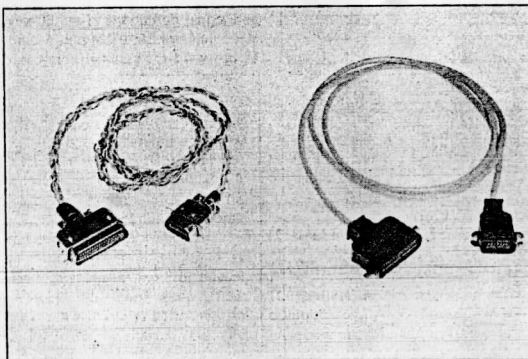
Listing 2. Suggestion for a test program to verify correct operation of the modified digitizer board.

```

10 OPEN"grp:"FOROUTPUTAS#1
20 ACOL=9:BCOL=8:COL=14
30 R1=8:R2=8:U1=4:U2=4
40 SCREEN2,2
50 COLOR15,4,4
60 CLS
70 LINE(32,0)-(32,176)
80 LINE(31,0)-(31,176)
90 LINESTEP(+0,+0)-STEP(+210,+0)
100 LINESTEP(+0,+0)-STEP(-210,+0)
110 FORY=155TO15STEP-20
120 PRESET(0,Y-2)
130 PRINT#1,MID$(STR$(155-Y),1);";";
140 LINE(31,Y)-(-36,Y+1)
150 NEXTY
160 GOSUB180
170 Q=0:GOTO160
180 FORA=1TO8
190 OUT0,A-1
200 H(A)=INP(0)/1.6
210 A$(A)=CHR$(A+48)
220 NEXT
230 FORN=56TO232STEP24
240 Y=168:X=N
250 Q=Q+1:H=H(Q)
260 A$=A$(Q)
270 IFNOT(HC=M(Q)-10RH)M(Q)+1)THEN320
280 PRESET(X-5,Y+24)
290 PRINT#1,RIGHT$(A$,2);
300 IFP=LANDM(Q)=>4THENGOSUB350ELSEGOSUB380
310 M(Q)=H
320 NEXTN
330 P=1
340 RETURN
350 LINE(X-1,Y-U1-H)-(X-R1,Y-U1),ACOL,BF
360 LINE(X,Y-U2-H)-(X-1+R2,Y-U1),BCOL,BF
370 GOTO400
380 LINE(X-1,Y-U1-H)-(X-R1,Y),ACOL,BF
390 LINE(X,Y-U2-H)-(X-1+R2,Y),BCOL,BF
400 LINE(X-R1,0)-(X-1+R2,Y-U1-H),3,BF
410 PRESET(X,Y-H)
420 DRAW" c=ccol;a2m=r1;,+u1;m=r2;,+u2; "
430 PRESET(X-1,Y-H-U1-U2)
440 DRAW" c=ccol;a0m=r1;,+u1;m=r2;,+u2; "
450 PAINT(X,Y-U1-H),CCOL
460 PRESET(X,Y)
470 DRAW" c=b;1+r1;u=u1;m=r1;,+u1; "
480 PAINTSTEP(-4,-1),0
490 PRESET(X,Y),0
500 DRAW" r=r2;u=u2;m=r2;,+u2; "
510 PAINT STEP(+4,-1),0
520 RETURN

```

Listing 3.
Graphic
presentation of
the input
signals to the
digitizer is
possible after
keying in this
program writ-
ten in Microsoft
Extended
BASIC



tion, however, may not be incorporated at all, so it is still sensible to switch off the computer before plugging in or removing any cartridges.

Testing

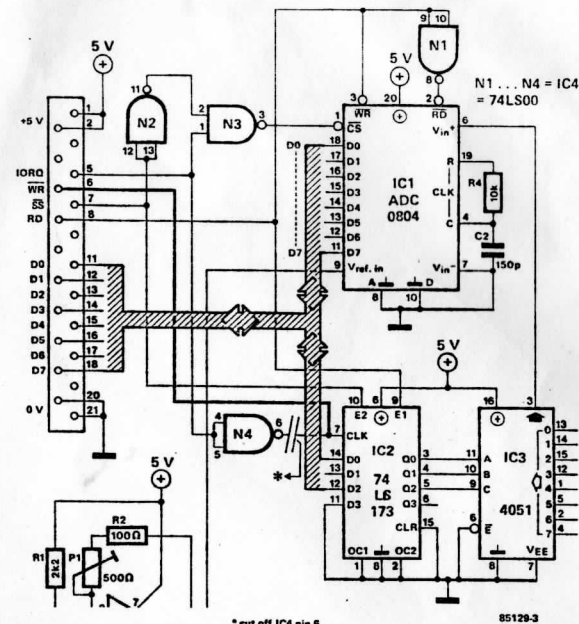
After completion of the wiring job using Table 1 as a guide, an initial test of the I/O bus may be performed. The digitizer and the 8-bit I/O must be removed for this purpose.

The Z80 processor features 256 I/O addresses, and every slot on the I/O bus takes four of them. With all switches on block S₂ closed, the computer will find the slots in the address range 0...15. To check the correct function of the modified address decoding part, two bits of every slot are hard-wired to its SS signal (pin 7), thus creating a byte to be read by the computer at the appropriate address. At slot 1, connector pins 17 and 18 are wired to pin 7. At slot 2, pins 15 and 16, at slot 3 pins 13 and 14 and, finally, pins 11 and 12 go to pin 7 of slot 4. Remember that the slot numbers refer to the new, reversed order, and that the computer may only read these addresses when the wires are present.

Once the computer addresses a slot, its SS signal goes low, together with the connected databits. This results in the following decimal values to be read from the slots: slot 1 = 63; slot 2 = 207; slot 3 = 243; and slot 4 = 252. Listing 1 shows a program to test all I/O locations from 0...255. It will skip any addresses reading 255, i.e. with all bits high. The range from 0...127 is reserved for the I/O bus, as MSX itself uses 128 and onwards. Locations 152 and 162, for instance, contain the VDP (Video Display Processor) and PSG (Programmable Sound Generator) status, respectively. In case the indicated decimal values are not read correctly by the program, some wiring error may have been made, or the jumpers b, d, and g are not yet in their correct positions.

If the first test-run is successful, the modified digitizer board may be plugged in and tested using program Listing 2. Fig. B shows how the digitizer is tested by connecting the wipers of eight preset potentiometers to the appropriate inputs. The potentiometers are connected between the +5 V supply and earth. Program Listing 2 will start the A/D converter eight times to subsequently read the converted values. Turning the presets will verify the correct function of the digitizer

3

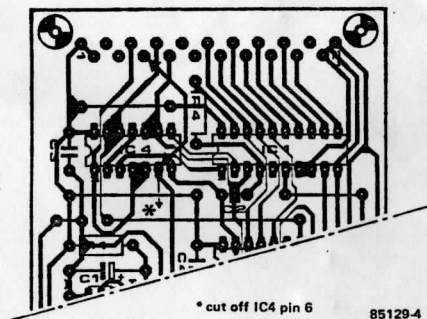


board. Because Listing 2 uses slot address 112, switches 12, 13, and 14, as indicated on the PCB, will have to be opened.
This program is also useful to test the

8-bit I/O board. All inputs are connected direct to the outputs by eight wires. The program will read back the sent values 0...7 as 255...248 because of the bit-inversion by the

ULN 2803. By changing A in line 20 to NOT A AND 255, the test program will print two neat columns of 0...7.

4



Practical example

The input signals to the digitizer are highly suitable for graphic presentation on the screen, as they are simultaneously visualized. Listing 3 suggests a program to implement this. The value at each input is shown as a vertical bar on the screen. A scale enables the values to be read off. Lines 20 and 30 of the program allow experiments with colours and bar sizes.

The vertical axis is drawn in lines 70 and 80, the horizontal one in lines 90 and 100. The scale is set up in lines 110...150, completing the basic layout of the screen. Line 170 is the central starting point in the program, the remainder of which consists of two subroutines. Lines 180...220 first read the input data of the digitizer before lines 230...340 put

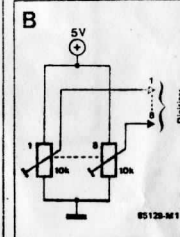
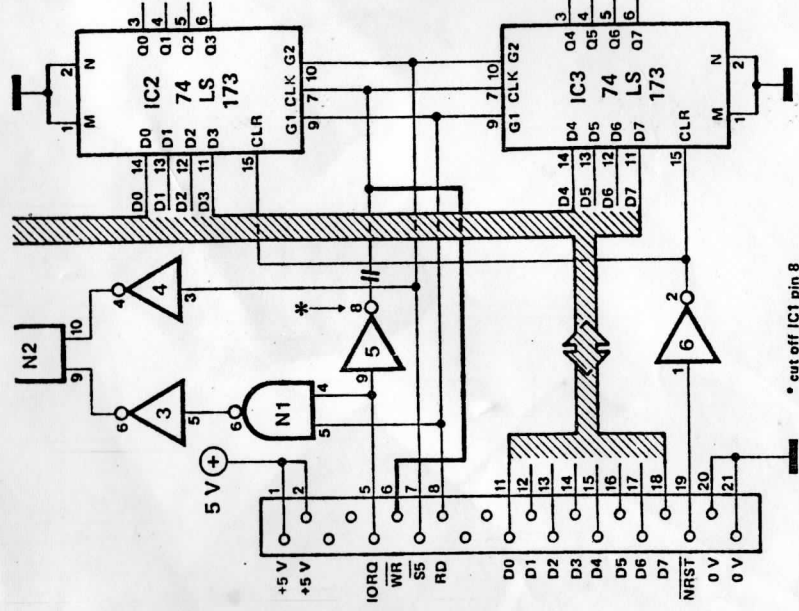


Fig. 3.
Modified circuit diagram of the digitizer board. One of the necessary alterations involves the removal of pin 6 from IC4.

Fig. 4. PCB of the digitizer, including the extra wire link.

Fig. 5. The heavy lines indicate the modifications to the 8-bit I/O board.

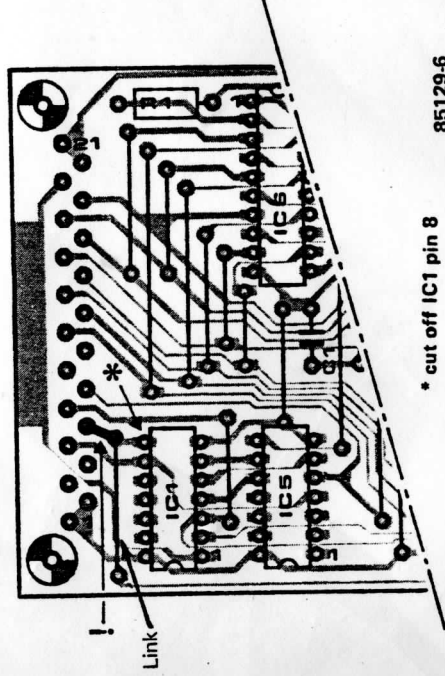
5



* cut off IC1 pin 8

Fig. 6. Modified PCB of the 8-bit I/O port. Just as with the digitizer, an IC pin has to be removed; in this case pin 8 of IC1.

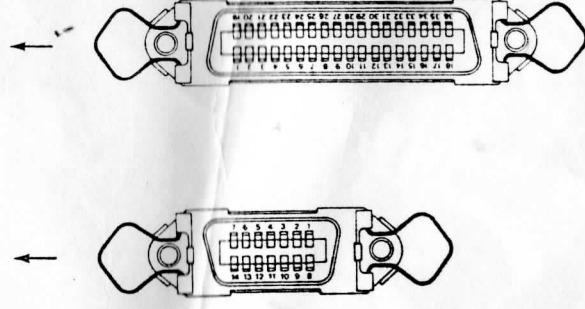
6



* cut off IC1 pin 8

85129-6

Fig. 7. It pays to make one's own printer cable. Eleven wires in a cable between two connectors do the job.



85129-7

them on the screen using another subroutine from line 350 onwards, which merely converts the data into coloured bars. To erase any previously higher values, line 400 puts a green part on top of every bar. As a finishing touch, lines 410...510 provide bevelled edges to the bars.

A printer cable

As soon as the computer starts to

grow into a so-called 'system', cables are required to connect peripheral equipment like a disc drive, a modem, or a printer. Considering the cost of a ready-made cable, it is certainly worthwhile to make one yourself for connecting a printer. Fig. 7 lists the necessary connections in the form of a table; the 14-way connector goes into the computer and the 36-way type (known as a Centronics connector) into the printer.

More interesting extensions of MSX computers will be featured in future issues of *Elektor Electronics*.